

```

1 <!DOCTYPE html>
2 <html lang="en">
3 <head>
4   <meta charset="UTF-8"/>
5   <meta name="description" content="Data Visualisation"/>
6   <meta name="keywords" content="HTML, CSS, D3"/>
7   <meta name="author" content="Anh Khoa"/>
8   <title>LAB 5.2 D3 Transitions</title>
9   <script src="https://d3js.org/d3.v7.min.js"></script>
10 </head>
11
12 <body>
13   <h1>The D3 Journey starts here...</h1>
14
15   <!-- Buttons for different transitions -->
16   <div style="margin-bottom: 20px;">
17     <button id="defaultTransition">Default Transition</button>
18     <button id="easeElastic">Elastic Transition</button>
19     <button id="easeCircle">Circle Transition</button>
20   </div>
21
22   <script>
23     var w = 500;
24     var h = 200;
25     var maxValue = 25;
26
27     var dataset = [14, 5, 26, 23, 9, 12, 18, 25];
28
29     var svg = d3.select("body")
30       .append("svg")
31       .attr("width", w)
32       .attr("height", h);
33
34     var xScale = d3.scaleBand()
35       .domain(d3.range(dataset.length))
36       .rangeRound([0, w])
37       .paddingInner(0.1);
38
39     var yScale = d3.scaleLinear()
40       .domain([0, maxValue])
41       .range([h, 0]);
42
43     // Initial bars
44     svg.selectAll("rect")
45       .data(dataset)
46       .enter()
47       .append("rect")
48       .attr("x", (d, i) => xScale(i))
49       .attr("y", d => yScale(d))
50       .attr("width", xScale.bandwidth())
51       .attr("height", d => h - yScale(d))
52       .attr("fill", "green");
53
54     // Initial labels
55     svg.selectAll("text")
56       .data(dataset)
57       .enter()
58       .append("text")
59       .text(d => d)
60       .attr("x", (d, i) => xScale(i) + xScale.bandwidth() / 2)
61       .attr("y", d => Math.max(yScale(d) - 5, 10))
62       .attr("text-anchor", "middle")
63       .attr("font-size", "12px")
64       .attr("fill", "black");
65
66     // Reusable chart update function
67     function updateChart(easeType) {
68       var numValues = 8;
69       dataset = d3.range(numValues).map(() => Math.floor(Math.random() * maxValue));
70       xScale.domain(d3.range(dataset.length));
71       yScale.domain([0, maxValue]);
72
73       const totalDuration = 1000;
74
75       // Update bars
76       svg.selectAll("rect")
77       .data(dataset)
78       .join("rect")
79       .transition()
80       .duration(totalDuration)
81       .delay((d, i) => i * (totalDuration / dataset.length))
82       .ease(easeType)
83       .attr("x", (d, i) => xScale(i))
84       .attr("y", d => yScale(d))
85       .attr("width", xScale.bandwidth())
86       .attr("height", d => h - yScale(d))
87       .attr("fill", "orange");
88
89       // Update labels
90       svg.selectAll("text")
91       .data(dataset)
92       .join("text")
93       .transition()
94       .duration(totalDuration)
95       .delay((d, i) => i * (totalDuration / dataset.length))
96       .ease(easeType)
97       .text(d => d)
98       .attr("x", (d, i) => xScale(i) + xScale.bandwidth() / 2)
99       .attr("y", d => Math.max(yScale(d) - 5, 10));
100     }
101
102     // Event listeners
103     d3.select("#defaultTransition").on("click", () => updateChart(d3.easeCubicInOut));
104     d3.select("#easeElastic").on("click", () => updateChart(d3.easeElasticOut));
105     d3.select("#easeCircle").on("click", () => updateChart(d3.easeCircleOut));
106   </script>
107
108   <footer style="color: grey; margin-top: 20px;">
109     COS30045 Data Visualisation<br>
110     Anh Khoa
111   </footer>
112 </body>
113 </html>

```